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A Two-Stage Diagnostic Framework for Time-Varying Structural Damage: Synergy of Modal Decomposition and Spatiotemporal Localization

Xiao-Mei Yang[1] | Xiao-Jun Yao[2] | Jian Zhang[3] | Zhenkun Li[4] | Ze-Long Sun[2]

[1]College of Civil Engineering, Fuzhou University, Fuzhou 350108, China | [2]School of Civil and Transportation Engineering, Hebei University of Technology, Tianjin 300401, China | [3]School of Civil Engineering, Tianjin Chengjian University, Tianjin 300192, China | [4]Department of Architecture, Built Environment and Construction Engineering, Politecnico di Milano, Milan 20133, Italy

Correspondence: Jian Zhang (zhangjian_dlut@163.com) | Zhenkun Li (zhenkun.li@polimi.it)

Received: 29 January 2026 | **Revised:** 14 September 2026 | **Accepted:** 18 September 2026

Academic Editor: Lin Chen

Keywords: Adaptive Chirp Mode Decomposition (ACMD) | modal decomposition | spatiotemporal localization | time-varying structural damage | two-stage diagnostic framework

ABSTRACT

Structural damage induced by material aging or long-term fatigue is a gradual process. The primary challenge in identifying such progressive damage lies in the fact that damage-induced changes in structural properties occur incrementally over time, with their effects simultaneously being manifested in both the temporal sequence and spatial distribution of vibration signals. These two aspects are intrinsically coupled and difficult to separate directly. To address this, this study proposes a two-stage diagnostic framework for time-varying structural damage. In the first stage of modal decomposition and tracking, a time–frequency ridge guidance mechanism is introduced to optimize the initial instantaneous frequency estimation for Adaptive Chirp Mode Decomposition (ACMD). This effectively mitigates the sensitivity to initial values, enabling robust decomposition of multiple intrinsic mode functions (IMFs) of the structure and their complete time-varying modal parameters. The second stage focuses on spatiotemporal localization: in the temporal domain, an Autoregressive Integrated Moving Average (ARIMA) model is established for each decoupled modal component. By leveraging the abrupt statistical changes in the statistical characteristics of the residual sequence, a Damage Index (DI) is constructed to sensitively capture the moment of damage occurrence. In the spatial domain, by integrating damage-sensitive time-varying frequency shifts and local mode shape distortions, a Damage Quantification (DQ) index along with its time-varying confidence boundary is developed, enabling accurate localization of the damage. The effectiveness of the proposed framework is validated through two numerical case studies including a time-varying stiffness multi-degree-of-freedom frame and a simply supported beam under moving loads, as well as an experimental test on a steel truss bridge subjected to ambient excitation. The results demonstrate that the framework not only identifies the precise timing of damage occurrence but also accurately localizes the damage position, providing a reliable solution for the sensing and preventive maintenance of time-varying structures.

1 | Introduction

Most structures are subject to the risk of deterioration or damage due to various external factors or internal defects encountered during service. These conditions can degrade structural performance and stability to varying degrees, potentially leading to functional failure or even collapse. The gradual degradation of structural systems can be regarded as a time-dependent dynamic

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process [1–3]. Therefore, the development of time-varying damage identification techniques is of critical importance. Damage identification objectives can generally be categorized into four progressive levels: detecting the existence of damages, locating damages, assessing damage severity, and predicting remaining service life [4, 5]. Most current research focuses on the first three levels. Signal processing techniques, including methods in the time domain, frequency domain, and time–frequency domain applied to field measurement data for damage identification, have been extensively studied by researchers [6, 7]. Time-domain methods often utilize time-series models and Kalman filtering algorithms [8, 9], while frequency-domain methods analyze transformed signals such as frequency response functions [10]. Time–frequency domain methods primarily rely on wavelet transforms or empirical mode decomposition (EMD) [11, 12].

Current progressive damage identification methods include approaches based on short-time invariant assumptions and time–frequency analysis. Mosavi et al. [13] utilized a vector autoregressive moving average model to analyze partitioned data within a moving short-time window, extracting damage properties through model parameters. The combination of recursive principal component analysis (PCA) with a time-varying autoregressive modeling method was employed for identifying progressive damage [14]. Singh and Sadhu introduced a time-varying filtering-based EMD method for progressive damage identification [15]. Tracking time-varying modal parameters plays a crucial role in pinpointing the timing and location of damage, as structural damage often correlates with changes in the physical parameters of the structure.

In recent years, the identification of time-varying modal parameters using time–frequency analysis techniques has garnered significant attention. The Hilbert–Huang transform method is a widely used approach for time–frequency analysis, particularly effective for processing signals in nonlinear and nonstationary processes [16, 17]. To enhance the performance of the Hilbert–Huang transform, factors such as ambiguities in intrinsic mode function (IMF) definitions, end effects in the Hilbert transform process, and the nonconvergence of the EMD algorithm need to be considered [18]. Dragomiretskiy [19] developed the variational mode decomposition (VMD) algorithm to address mode mixing issues by identifying orthogonal modes within each IMF, and has been successfully applied in decomposing nonstationary signals and identifying instantaneous frequencies [20–22].

Although VMD has shown efficacy in modal identification, it does not directly estimate time-varying features such as instantaneous frequency, limiting its effectiveness in capturing the dynamic nature of structures. To address this limitation, Chen [23] introduced the Adaptive Chirp Mode Pursuit (ACMP), which captures signal modes one by one in a recursive framework. Adaptive Chirp Mode Decomposition (ACMD) is a signal decomposition method for nonstationary signals based on the VMD framework [24]. ACMD autonomously estimates each signal component without requiring a predefined number of components, offering improved handling of signals with overlapping frequencies compared to VMD. Additionally, ACMD allows for the simultaneous extraction of IMFs and instantaneous modal parameters, facilitating damage identification in dynamic structures. The ACMD algorithm has been applied in various fields including mechanical fault diagnosis, oscillatory signal parameter identification, and biomedical signal processing. However, its potential in the domain of time-varying structural damage identification remains underexplored [24–27].

The decomposition performance of the ACMD largely depends on the accuracy of the initial instantaneous frequency estimation. In complex structural health monitoring environments, the absence of a reliable initial instantaneous frequency guidance mechanism tends to cause modal extraction errors or omissions during ACMD implementation, potentially leading to missed detections or false alarms in subsequent damage identification [24, 25]. Most existing methods do not dynamically adjust the initial values based on the time-varying characteristics of the signals. Consequently, users require considerable prior knowledge or debugging experience, while the method itself lacks adaptive tracking capability. This limitation hampers the widespread application of ACMD in complex scenarios for damage identification. Furthermore, the outputs of traditional ACMD are limited to IMFs and their parameters, such as instantaneous frequency and amplitude, without establishing clear diagnostic correlations with the specific timing and spatial location of damage. This lack of a systematic analytical framework from modal decomposition to spatiotemporal damage localization restricts its practical utility.

To address the initial value sensitivity and manual parameter tuning of traditional ACMD, recent data-driven deep learning strategies for modal identification have eliminated the requirement for manually preset initial instantaneous frequencies and reduced reliance on empirical priors [28–30]. Nevertheless, deep learning–based pipelines typically involve substantial computational cost, which poses challenges for real-time on-site structural health monitoring. In addition, existing studies predominantly focus on modal decomposition alone, without providing quantitative metrics for temporal damage detection or spatial damage localization, which limits their direct deployment for diagnosing progressive degradation in time-varying structures.

Despite these advances, identifying progressive structural damage, characterized by subtle, time-varying property changes, remains a significant challenge in structural health monitoring due to the intricate coupling of temporal and spatial damage manifestations. This study addresses these limitations by proposing a novel two-stage diagnostic framework for robust identification and precise localization of such damage based on time-varying modal parameters. In the first stage, an enhanced ACMD method is proposed, which integrates a novel time–frequency ridge guidance mechanism to overcome initial value sensitivity, enabling accurate extraction of a complete set of time-varying modal parameters. Building upon this, the second stage introduces a spatiotemporal collaborative localization strategy. For temporal localization of damage onset, ARIMA models are fitted to individual time-varying modal components, and the resulting residuals are leveraged to construct a sensitive Damage Index (DI). For spatial localization, a novel Damage Quantification (DQ) index that synergistically combines time-varying frequency shifts and local mode shape distortions is developed, together with a time-varying confidence boundary (CB) for robust damage position identification. This integrated

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approach offers a comprehensive and reliable solution for the sensing and preventive maintenance of structures undergoing gradual degradation.

The rest of this article is organized as follows. The time-varying structural vibration signals are analyzed and the ACMD and ARIMA models are reviewed in Section 2. Next, the proposed two-stage diagnostic framework is introduced. The damage indices DI and DQ for temporal and spatial localization are discussed in Section 3. In Sections 4 and 5, the effectiveness of the proposed framework is demonstrated through simulation and experimental studies. Finally, the concluding remarks are presented in Section 6.

2 | Theoretical Foundations of the Algorithms

2.1 | Introduction to the ACMD Method

The ACMD method [24] operates on the principle that any multi-degree-of-freedom vibration signal can be decomposed into N IMFs, as described in equation (1). Each IMF captures distinct local time–frequency characteristics of the signal.

$$f(t)=\sum_{i=1}^{N}u_i(t)=\sum_{i=1}^{N}A_i(t)\cos\left(2\pi\int_0^t\omega_i(\tau)d\tau+\psi_i\right), \tag{1}$$

where $f(t)$ is a vibration signal to be decomposed; $u_i(t)$ represents the i th decomposed IMF; and $A_i(t)$, $\omega_i(\tau)$, and ψ_i represent the i th instantaneous amplitude, frequency, and phase, respectively. By employing signal demodulation methods, equation (1) can be rewritten as

$$f(t)=\sum_{i=1}^{N}\left\{\alpha_i(t)\cos\left(2\pi\int_0^t\widetilde{\omega}_i(\tau)d\tau\right)+\beta_i(t)\sin\left(2\pi\int_0^t\widetilde{\omega}_i(\tau)d\tau\right)\right\}, \tag{2}$$

where

$$\begin{cases}\alpha_i(t)=A_i\cos\left\{2\pi\int_0^t\left[\omega_i(\tau)-\widetilde{\omega}_i(\tau)\right]d\tau+\psi_i\right\},\\ \beta_i(t)=-A_i\sin\left\{2\pi\int_0^t\left[\omega_i(\tau)-\widetilde{\omega}_i(\tau)\right]d\tau+\psi_i\right\},\end{cases} \tag{3}$$

where $\cos(2\pi\int_0^t\widetilde{\omega}_i(\tau)d\tau)$ and $\sin(2\pi\int_0^t\widetilde{\omega}_i(\tau)d\tau)$ are orthogonal demodulation operators; and $\widetilde{\omega}_i$ is the frequency of the demodulation operators $\alpha_i(t)$ and $\beta_i(t)$. The relationship between the instantaneous amplitude in equation (1) and the demodulated operators in equation (3) is

$$A_i(t)=\sqrt{\alpha_i^2(t)+\beta_i^2(t)}. \tag{4}$$

The target IMFs and their instantaneous frequencies can be calculated by solving the following equations:

$$\begin{cases}\min\limits_{\alpha_i,\beta_i,\widetilde{\omega}_i}\left\{\sum_{i=1}^{N}\left[\left\|\alpha_i^*(t)\right\|_2^2+\left\|\beta_i^*(t)\right\|_2^2+\upsilon\|f(t)-u_i(t)\|_2^2\right]\right\},\\ \text{s.t. } u_i(t)=\alpha_i(t)\cos\left(2\pi\int_0^t\widetilde{\omega}_i(\tau)d\tau\right)+\beta_i(t)\sin\left(2\pi\int_0^t\widetilde{\omega}_i(\tau)d\tau\right),\end{cases} \tag{5}$$

where $\|\cdot\|_2^2$ and υ represent the Euclidean norm and penalty parameter.

Considering that the vibration signal $f(t)$ is discretely sampled as $\mathbf{f}=[f(t_0),\ldots,f(t_{M-1})]^T$, and the number of samples is M, equation (5) takes the discrete form

$$\min_{u_i,\omega_i}\left\{\|\mathbf{\Theta}\mathbf{v}_i\|_2^2+\upsilon\|\mathbf{f}-\mathbf{G}_i\mathbf{v}_i\|_2^2\right\}, \tag{6}$$

where $\mathbf{\Theta}$, $\boldsymbol{\nu}_i$, and $\mathbf{G}_i$ are

$$\mathbf{\Theta}=\begin{pmatrix}\mathbf{\Omega} & 0\\ 0 & \mathbf{\Omega}\end{pmatrix}. \tag{7}$$

$$\mathbf{v}_i=\left[\alpha_i(t_0)\cdots\alpha_i(t_{M-1})\beta_i(t_0)\beta_i(t_1)\cdots\beta_i(t_{M-1})\right]^T. \tag{8}$$

$$\mathbf{G}_i=\mathrm{diag}[\cos(\kappa_i(t_0))\cdots\cos(\kappa_i(t_{M-1}))\sin(\kappa_i(t_0))\sin(\kappa_i(t_1))\cdots\sin(\kappa_i(t_{M-1}))], \tag{9}$$

where $\mathbf{\Omega}$ denotes a second-order difference matrix [20], and $\kappa_i(t)$ is

$$\kappa_i(t)=2\pi\int_0^t\widetilde{\omega}_i(t)dt. \tag{10}$$

The demodulated operators are iteratively updated as

$$\mathbf{v}_i^j=\arg\min_{\upsilon}\left\{\|\mathbf{\Theta}\mathbf{v}_i\|_2^2+\upsilon\left\|\mathbf{f}-\mathbf{G}_i^j\mathbf{v}_i\right\|_2^2\right\}=\left[\frac{1}{\upsilon}\mathbf{\Theta}^T\mathbf{\Theta}+\left(\mathbf{G}_i^j\right)^T\left(\mathbf{G}_i^j\right)\right]^{-1}\left(\mathbf{G}_i^j\right)^T\mathbf{f}. \tag{11}$$

The IMF is correspondingly obtained as

$$\mathbf{u}_i^j=\mathbf{G}_i^j\mathbf{v}_i^j. \tag{12}$$

The difference ξ_i^j between two consecutive iterations is calculated by

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$$\xi_i^j = \frac{\left\| \boldsymbol{u}_i^j - \boldsymbol{u}_i^{j-1} \right\|_2^2}{\left\| \boldsymbol{u}_i^j \right\|_2^2} < \varepsilon. \tag{13}$$

The iteration stops when the difference ξ_i^j is less than the preset convergence criterion ε.

2.2 | Theory of Time-Series Model for Damage Identification

The time-series model is a statistical analysis method that attempts to utilize mathematical models to fit temporal data, thereby extracting useful information from the model. Structural damage identification can be achieved by identifying changes in the time-domain response signals. The Autoregressive Integrated Moving Average (ARIMA) model aims to describe the current value of a variable based on its linear relationship with past values [31]. The ARIMA model can be decomposed into two parts: the first part involves differencing the time series to achieve stationarity; the second part entails fitting an Autoregressive Moving Average (ARMA) model to the differenced time series, which appears stationary. The general form of the ARMA model is

$$X_t = \sum_{i=1}^{p} a_i X_{t-i} - \sum_{j=1}^{q} b_i e_{t-j} + e_t, \tag{14}$$

where p is the autoregressive order, a_i is the autoregressive coefficient, X_t is the input time series, q is the moving average order, b_i is the moving average coefficient, and e_t is the model residual.

For any value on the time series, the lag operator is defined as

$$L(X_t) = X_{t-1}. \tag{15}$$

To this end, higher-order lag operators can be defined as

$$L^i(X_t) = X_{t-i}. \tag{16}$$

Thus, the representation of the ARMA model using lag operators becomes

$$\left(1 - \sum_{i=1}^{p} a_i L^i\right) X_t = \left(1 - \sum_{i=1}^{q} b_i L^i\right) e_t. \tag{17}$$

Denoting $a(L) = (1 - \sum_{i=1}^{p} a_i L^i)$, $b(L) = (1 - \sum_{i=1}^{q} b_i L^i)$, it can be derived as

$$X_t = \frac{b(L)}{a(L)} e_t. \tag{18}$$

From the above equations, the ARMA model can be transformed into a transfer function form. The model assumes that the random excitation from the external environment on the system is white noise. The transfer function form indicates that the ARMA model is capable of simulating the impact of the white noise input on system response, thereby establishing an equivalent relationship with the dynamic behavior of the actual system in time-series analysis. Therefore, if the inputs and systems are the same, the ARMA residuals of their outputs will be small. The ARMA models will generate larger residuals for different systems [31].

The differencing method is commonly used to convert the nonstationary time series into a stationary one, which involves a differencing procedure applied to the nonstationary data. An ARMA model with a differencing process can be referred to as an ARIMA model. The ARIMA model can be specifically expressed as follows [32].

$$\left(1 - \sum_{i=1}^{p} a_i L^i\right)(1 - L)^d X_t = \left(1 - \sum_{i=1}^{q} b_i L^i\right) e_t, \tag{19}$$

where d is the differencing order. Time-series models are capable of detecting changes in the statistical characteristics of time-series data. When damage occurs to a structure, it typically results in abrupt changes in the data or alterations in the dynamic properties of the system, leading to significant discrepancies between the monitored data and the established time-series model. Based on this characteristic, damage indicators based on residuals can be constructed to identify abrupt structural damage.

The ARIMA-based damage detection strategy presented above relies on two fundamental assumptions that are standard in output-only operational modal analysis. First, the unmeasured excitation is assumed to be a stationary, zero-mean, broadband random process (ideally Gaussian white noise). This ensures that the output spectrum preserves the same modal information as the system's frequency response function, enabling reliable extraction of structural modes. Second, the structural behavior is assumed to be linear and time-invariant within each analysis window, meaning that future responses depend linearly on past observations, which is a condition captured by the autoregressive formulation of the ARIMA model. These assumptions are reasonably satisfied for typical ambient vibration scenarios such as wind, traffic, and microtremors, where the excitation is broadband and the structure remains within its elastic range. However, it is important to acknowledge that the method is not intended for extreme nonstationary events such as strong earthquakes, where the excitation spectrum is highly colored and time-varying, and the structural response may exhibit significant nonlinearity. In such cases, the identified modal parameters may be biased, and the damage indicators may lose their quantitative interpretability. The applicability of the proposed framework to such extreme scenarios is, therefore, not guaranteed and is reserved for future research.

3 | A Two-Stage Synergistic Diagnostic Framework for Time-Varying Structural Damage

The proposed framework builds upon several established techniques, including ACMD [24], ARIMA models [31], and sliding-window PCA [33], but introduces three targeted innovations for progressive time-varying structural damage diagnosis. First, a time–frequency ridge guidance mechanism is developed to automatically generate initial instantaneous frequencies for ACMD, overcoming its inherent sensitivity to manually preset starting values. Second, ARIMA models are separately fitted to each decoupled modal component, and model residuals are leveraged to construct a sensitive DI for accurate temporal localization of damage onset. Third, a novel DQ index integrated with time-varying confidence boundaries is formulated, which synergistically combines time-varying frequency shifts and local mode shape distortions to realize

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reliable spatial damage identification. The following subsections elaborate the principles and implementation of each improved module.

3.1 | ACMD Initialization and Parameter Extraction via Time–Frequency Ridge Guidance

The ACMD algorithm requires initial instantaneous frequencies as input to extract specific frequency orders. Accurate determination of these initial frequencies is crucial for precise identification of instantaneous modal parameters. To enhance decomposition performance, this study introduces a sliding-window AR power spectrum estimation method for initial frequency extraction, implemented through three key steps.

i. Window configuration: An appropriate window function $w(n)$ of length N is selected to segment the vibration response data $f(n)$ as

$$x(n) = f(n) \times w(n). \tag{20}$$

The selected window function should be able to reduce spectral leakage at the window boundaries.

ii. AR power spectrum calculation: Sliding the window and performing AR power spectrum estimation on the local dataset within the window. Using the window function $w(n)$ to divide the vibration data $f(n)$ of length L into $((L-N)/d)+1$ segments, the data length of each segment is N. Partial overlap length of the window function is $N-d$. The AR model and AR power spectrum of each time segment are expressed as

$$x(n) = -\sum_{i=1}^{p} a_i x(n-i) + e(n). \tag{21}$$

$$P_{XX}\left(e^{j\omega}\right) = \frac{\sigma^2}{\left|1+\sum_{i=1}^{p} a_i e^{-j\omega i}\right|^2}, \tag{22}$$

where $P_{XX}(e^{j\omega})$ is the AR power spectrum for each segment.

iii. Initial frequency determination: The time–frequency distribution $TF(t,\omega)$ of the vibration signal is constructed using the AR power spectrum of $((L-N)/(N-Q))+1$ segments. The local maximum value $(t_{\max}, \omega_{\max})$ in the time-frequency distribution is detected as the starting point for ridge line extraction. Subsequently, the principle of frequency continuity is considered by selecting the frequency point with the highest-energy frequency point at adjacent time instants, namely, the left point $(t_{\max-1}, \omega_{\max-1}) = \max TF(t_{\max-1}, \omega)$ and the right point $(t_{\max+1}, \omega_{\max+1}) = \max(t_{\max+1}, \omega)$, where $\omega \in [\omega_{\max} - \Delta\omega, \omega_{\max} + \Delta\omega]$. $\Delta\omega$ is the maximum allowable frequency variation threshold. In this case, the highest-energy time-frequency ridge is obtained, and the ridge line frequency $\widetilde{\omega}(t)$ can be taken as the initial instantaneous frequency.

Building upon the above initial frequency determination method, the ACMD algorithm sequentially extracts each mode in descending order of modal energy magnitude, as follows.

Step (1): The process begins by obtaining the time–frequency distribution of the response signal through sliding-window AR power spectrum estimation.

Step (2): The highest-energy time–frequency ridge is then identified to provide the initial instantaneous frequency.

Step (3): The initial frequency serves as input to the ACMD algorithm, which extracts both the corresponding IMF and its instantaneous frequency. Following extraction, the obtained IMF is removed from the original signal, leaving a residual signal for subsequent processing.

Step (4): The residual signal undergoes the same procedure for further ridge line and IMF extraction. The iterations continue until all IMFs are sequentially extracted.

Unlike VMD, ACMD does not require predetermining the number of IMFs prior to decomposition. Furthermore, ACMD extracts IMFs in descending order of energy, enhancing the identification accuracy of low-energy modes. Additionally, the algorithm can simultaneously and accurately extract both IMFs and their corresponding instantaneous frequencies.

Notably, the ridge detection stage only requires a moderately resolved time–frequency representation, as the subsequent iterative refinement of ACMD is robust to minor deviations of extracted ridges. Consequently, neither ultrahigh spectrogram resolution nor extra-long data records are required; a moderate data length is sufficient to capture distinguishable energy ridges, and slight blurring of time–frequency features can be effectively compensated during ACMD iterations.

3.2 | Identification of Time-Varying Mode Shapes Using Sliding-Window PCA

Under the assumption that mode shapes remain stationary within short-time intervals, time-varying mode shapes can be identified using a sliding-window PCA approach. The method works by transforming multisensor response data into principal components (linearly uncorrelated variables capturing maximum variance [33]) and expressing structural responses as linear combinations of these mode shapes. The detailed implementation steps are as follows:

i. Window processing: For a discretely sampled IMF series $u_{ji}(t)$ of length L, apply a window function of length N with step size Δ to segment the data. Within each window, mode shapes are considered time-invariant for PCA identification.

There is a clear trade-off in selecting the window length N between temporal resolution and statistical stability. The minimum valid window size must satisfy two quantitative criteria: first, the window contains at least 5–10 complete vibration cycles of the structure's lowest mode of interest; second, the total number of samples exceeds 2-3 times the number of deployed sensors to construct a full-rank covariance matrix for reliable PCA decomposition. Windows shorter than this threshold lead to severe statistical fluctuations, spurious modal components and distorted mode shapes due to insufficient data. Conversely, excessively long windows violate the local quasistationary assumption and smooth out damage-sensitive time-varying signatures. For nonstationary events, the window length should be chosen as short as practically

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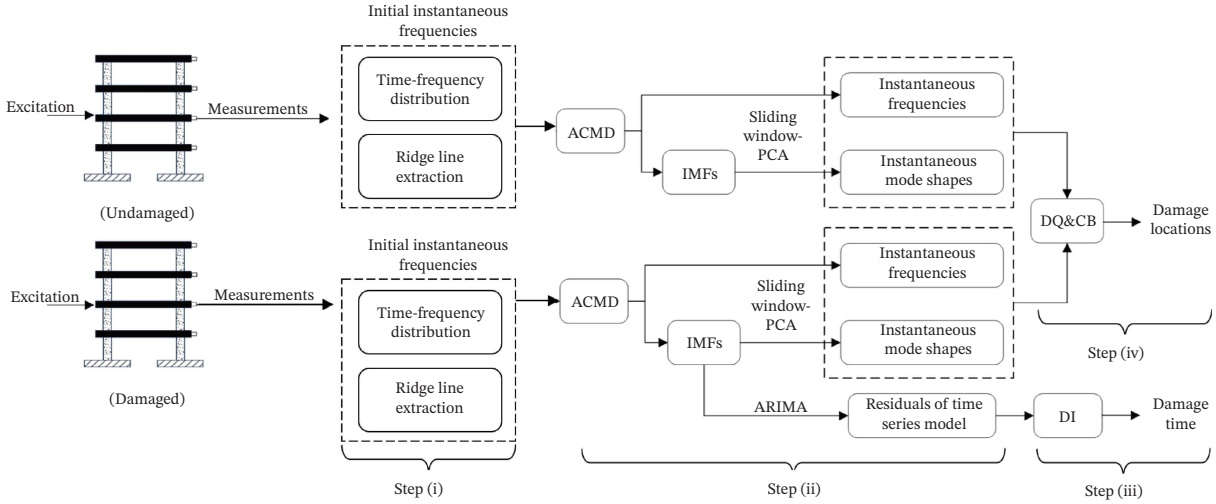


FIGURE 1 | Flowchart of the proposed damage identification method.

allowed by frequency resolution requirements to capture rapid modal changes.

ii. Data standardization: For the i th IMF vector $\mathbf{u}_i = \left[u_{1i} \;\; u_{2i} \;\; \cdots \;\; u_{Si} \right]^T$ across all S sensors and within a window, standardize the windowed data $\mathbf{u}_i$ to obtain the vector $\widehat{\mathbf{u}}_i$ with zero mean and unit variance.

iii. Mode shape identification: Perform singular value decomposition (SVD) on the covariance matrix of the standardized vector using the following equation.

$$\mathbf{R}_i(t) = E\left(\widehat{\mathbf{u}}_i(t)\widehat{\mathbf{u}}_i(t)^T\right) = \mathbf{P}(t)\mathbf{\Lambda}(t)\mathbf{P}(t)^T, \tag{23}$$

where $\mathbf{R}$ is the covariance matrix of the standardized vector; the first column of the left singular vector matrix $\mathbf{P}(t)$ represents the estimated i-th mode shape vector.

This quasistationary approximation holds well for normal ambient vibration scenarios with slow and mild stiffness degradation. However, the assumption is no longer valid under strong seismic excitation that triggers abrupt damage inside a single sliding window. In such cases, each window contains mixed predamage and postdamage modal information, which contaminates PCA feature extraction, distorts the spatial distribution of the DQ index, and degrades the accuracy of spatial damage localization. Extending the windowed PCA strategy to cope with intrawindow sudden modal variations induced by extreme loads will be investigated in future work.

3.3 | Construction of Dual Indices for Spatiotemporal Damage Localization

3.3.1 | ARIMA Residual-Based DI for Temporal Localization

The structural damage onset time is denoted by t_0, and the n-th IMF of the j-th vibration signal varying with time is expressed as $u_{jn} = [u_{jn}(1), \ldots, u_{jn}(t_0), \ldots, u_{jn}(t)]$. The ARIMA model residual at damage time t_0 is computed as

$$e_{jn}(t_0) = \widetilde{u}_{jn}(t_0) - \sum_{i=1}^{p} a_i \widetilde{u}_{jn}(t_0 - i) + \sum_{k=1}^{q} b_k e_{jn}(t_0 - k). \tag{24}$$

For time at $t < t_0$, IMF values represent undamaged structural behavior. Because the IMF values corresponding to undamaged structural behavior cannot properly characterize the damaged state at time $t \geq t_0$, residual values exhibit a significant discontinuity at time t_0. Therefore, the time corresponding to structural damage can be identified by analyzing the variation of the ARIMA model residuals.

A DI can be constructed using the residual sequences. By identifying the significant changes of the DI sequence, the moment of structural damage can be determined. The expression of DI at time t is

$$\mathrm{DI}(t) = \frac{\sum_{j=1}^{S}\sum_{i=1}^{N} e_{ji}(t)}{S \cdot N}, \tag{25}$$

where S is the total number of sensors.

3.3.2 | Frequency- and Mode Shape–Integrated DQ Index for Spatial Localization

To characterize structural damage progression as a time-varying nonstationary process, a DQ index with respect to the position of the s th sensor is constructed using time-varying modal parameters, as

$$\mathrm{DQ}_s(t) = 1 - \frac{\sum_{i=1}^{N}\left[\left((\varphi_{si}(t))_{\mathrm{ref}}\right)/\left((\omega_i(t))_{\mathrm{ref}}\right)\right]^2}{\sum_{i=1}^{N}\left[(\varphi_{si}(t))/(\omega_i(t))\right]^2}, \tag{26}$$

where φ_{si} represents the i th mode shape value at the s-th sensor, and ω_i represents the frequency of the i th mode; the subscript "ref" represents the reference status. The severity of abrupt structural damage can be quantified by the DQ value, which is a DI ranging from 0 (*undamaged*) to 1 (*completely damaged*).

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TABLE 1 | Hyperparameter configuration and selection rules for the proposed framework.

Module	Parameter	Value (s)	Selection method/rationale
Sliding window	Window type	Hann/Hamming	Standard choices for spectral analysis; providing smooth tapering to reduce spectral leakage
	Window length	512/1024 samples (specified per example)	Chosen to balance frequency resolution and time resolution; at least 5–10 times the fundamental period of the structure
	Overlap ratio	50%/75%	50% is standard; 75% used for cases requiring finer time tracking
AR spectrum estimation	AR model order	40/60	Determined by minimizing the Akaike Information Criterion (AIC) over a search range (10, 100) on the baseline dataset
ARIMA (p, d, q)	Order selection	Determined automatically per sensor-mode pair	Implemented using the auto.arima procedure, which searches over a grid of (p, d, q) values and selects the model that minimizes AICc; maximum orders were limited to $p \leq 5$, $d \leq 2$, $q \leq 5$
	Typical selected orders	$p \in [1, 3]$, $d \in [0, 1]$, $q \in [0, 2]$	Confirmed that selected orders are consistent with the expected low-order dynamics of modal feature time series
ACMD	Convergence tolerance (ε)	1×10^{-6}	Standard setting; verified that stricter tolerances (10^{-8}) produced negligible differences in extracted IFs ($< 0.1\%$)
	Maximum iterations	200	A maximum of 200 iterations is sufficient for convergence in all tested cases; early stopping typically occurred after 30–80 iterations
	Bandwidth parameter (α)	1×10^{-3} to 1×10^{-4}	Estimated from the ridge width in the time–frequency representation, with smaller α values assigned to finer modes
	Ridge extraction threshold	Top 5%–10% of TFR amplitude	Automatically determined from the time–frequency representation histogram; verified by visual inspection of extracted ridges

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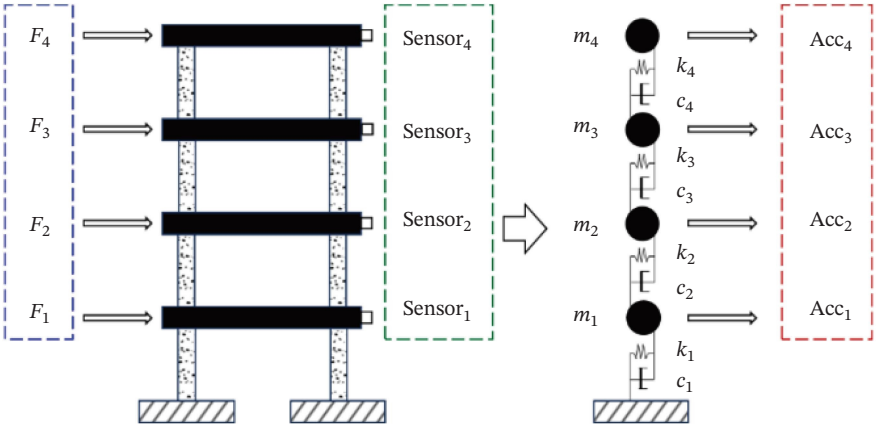


FIGURE 2 | The four-story frame structure.

3.3.3 | Construction of Time-Varying CB for Damage Localization

For reliable damage localization in time-varying structures, a CB is defined as

$$\text{CB} = E[\text{DQ}] + \frac{\left(t_{S-1,(1/2)+(\beta/2)}\right)\sqrt{E\left[(\text{DQ}-E[\text{DQ}])^2\right]}}{\sqrt{S}}, \quad (27)$$

where $t_{S-1,(1/2)+(\beta/2)}$ represents the percentile of the t-distribution corresponding to confidence level $1/2+\beta/2$; $S-1$ is the degree of freedom; $E[\text{DQ}]$ represents the expected value of DQ values within each time window; and $\sqrt{E[(\text{DQ}-E[\text{DQ}])^2]}$ represents the estimate of variability for DQ values within each time window. Assuming the baseline reference data follow a t-distribution, the mean and standard deviation of the baseline reference data can be used to estimate $E[\text{DQ}]$ and $\sqrt{E[(\text{DQ}-E[\text{DQ}])^2]}$ [34]. A position can be determined as the specific location of progressive damage when the DQ value consistently exceeds the CB.

3.4 | Implementation Flow of the Framework

The specific steps for the proposed damage identification method are summarized in Figure 1 and described in the following.

i. The initial instantaneous frequencies are obtained by extracting time–frequency ridge lines of the sliding-window AR power spectrum in equation (22);

ii. The measured responses are decomposed by the ACMD algorithm with the initial instantaneous frequencies as inputs to obtain the IMFs and instantaneous frequencies according to equation (5); the instantaneous mode shapes are obtained by performing sliding-window PCA on the IMFs;

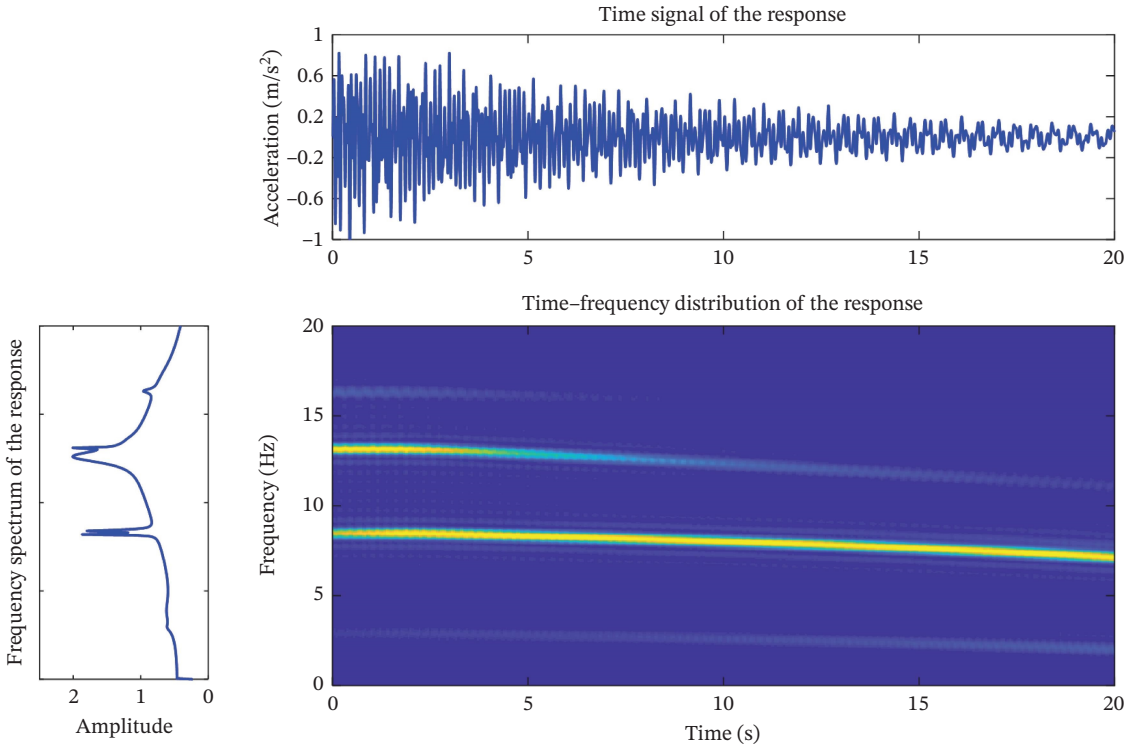


FIGURE 3 | Time–frequency distribution of the response of the first floor.

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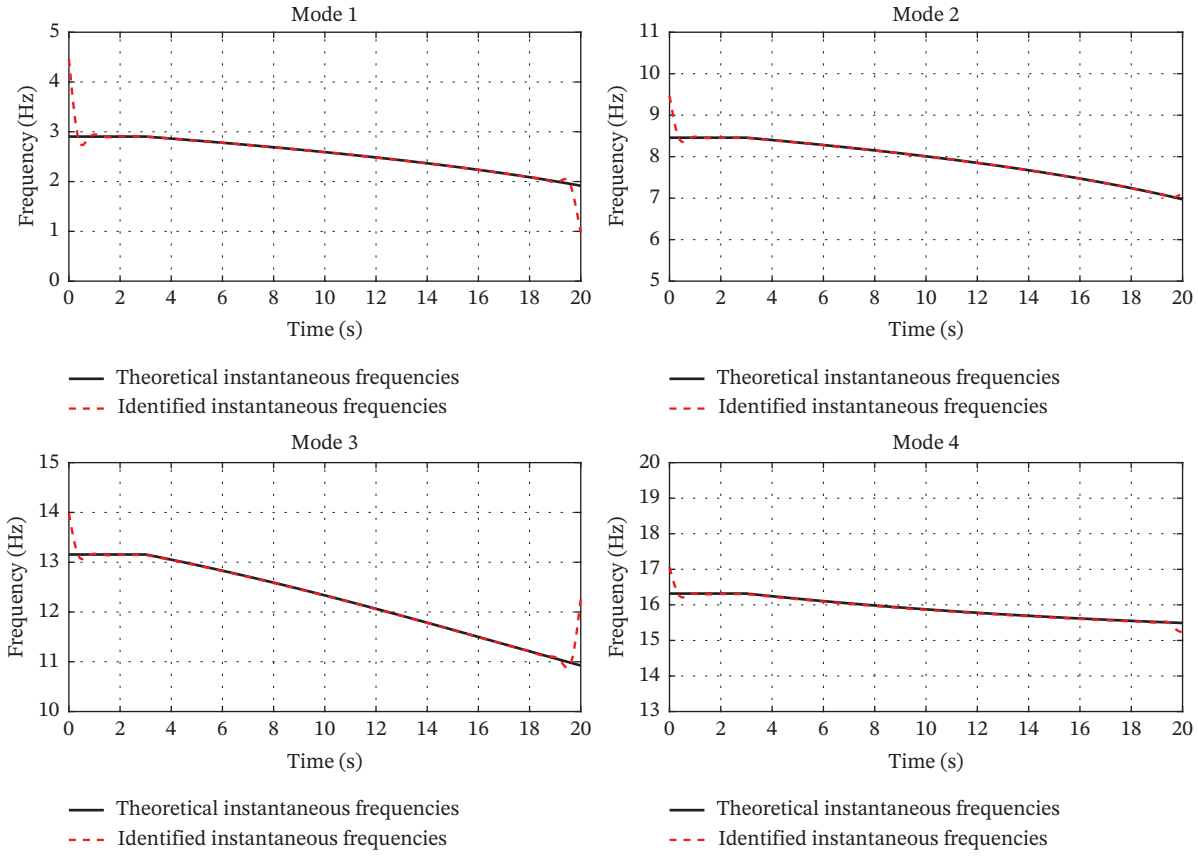


FIGURE 4 | Identified instantaneous frequencies based on ACMD.

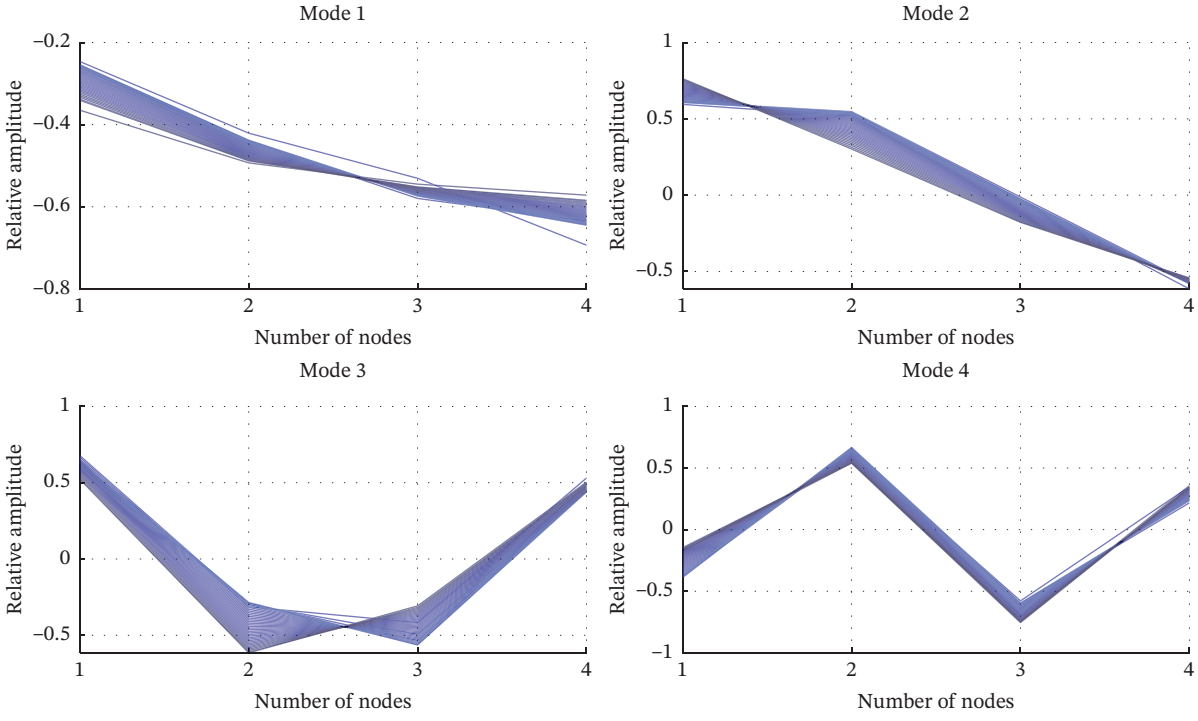


FIGURE 5 | Identified time-varying mode shapes.

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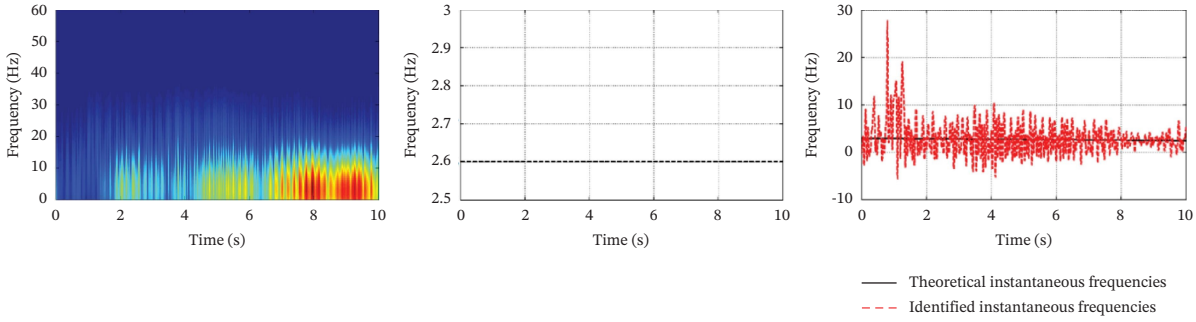


FIGURE 6 | Identified instantaneous frequencies by not well-chosen initial frequencies. (a) Time–frequency distribution by STFT. (b) Extracted initial instantaneous frequencies. (c) Identified instantaneous frequencies by ACMD.

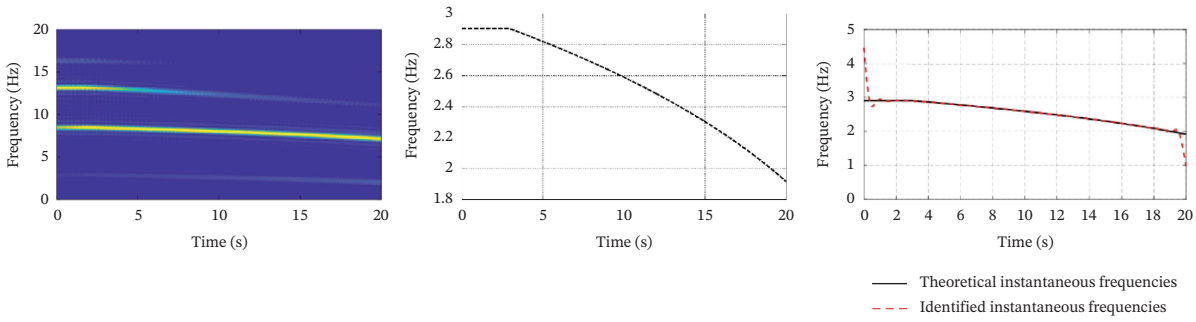


FIGURE 7 | Identified instantaneous frequencies by well-chosen initial frequencies. (a) Time–frequency distribution by the proposed method. (b) Extracted initial instantaneous frequencies. (c) Identified instantaneous frequencies by ACMD.

iii. The extracted IMFs are fitted using ARIMA model, and the residuals of the model are extracted. The DI is then constructed based on equation (25), with the occurrence of damage being determined by the abrupt changes in the DI values.

iv. The DI DQ and the time-varying CB are, respectively, constructed based on equations (26) and (27) to locate damage positions in time-varying structures.

Table 1 summarizes all hyperparameters adopted in the proposed two-stage framework, including the value ranges and corresponding selection rationales for sliding window processing, AR spectrum estimation, ARIMA order determination, and ACMD modal decomposition. Case-specific adjustable parameters are elaborated separately in each validation subsection.

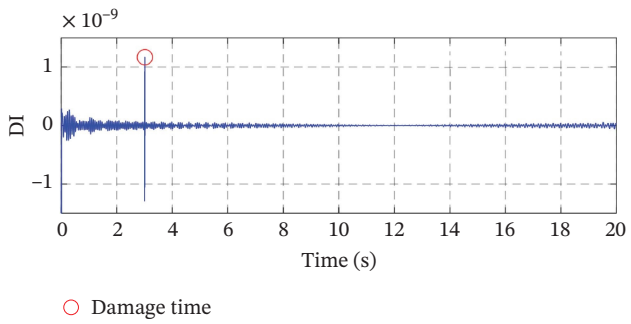


FIGURE 8 | Damage time identification results based on the damage index DI.

4 | Numerical Verification of Damage Identification in Time-Varying Structures

This section systematically validates the performance of the proposed two-stage diagnostic framework for identifying damage in time-varying structures through two representative numerical case studies. The first case aims to verify the framework's capability for spatiotemporal localization of damage in a frame structure, while the second case further validates its effectiveness in tracking damage evolution in a simply supported beam under the complex scenario of coupled moving loads and progressive damage.

4.1 | Simulation of a Time-Varying Four-Degree-of-Freedom (4-DOF) Frame Structure

4.1.1 | Simulation Model and Damage Scenario Setup

The effectiveness of the proposed damage detection framework is numerically verified through a 4-DOF frame structure exhibiting time-varying dynamic properties. Figure 2 illustrates the simulated structure. The symbols $F_1 \sim F_4$ represent the excitations applied to the structure, respectively. The masses of the structure are $m_1 = m_2 = m_3 = m_4 = 4\,\text{kg}$. The symbols k_1 and k_2 are time-varying stiffness, where $k_1 = 1 \times 10^4 N/m$ and $k_2 = 1.2 \times 10^4 N/m$ for time $t \le 3\,\text{s}$, while $k_1 = (1 - 0.04(t-3)) \times 10^4 N/m$ and $k_2 = (1.2 - 0.03(t-3)) \times 10^4 N/m$ for time $t > 3\,\text{s}$. The symbols k_3 and k_4 are time-invariant stiffness, and both of them are $1.2 \times 10^4 N/m$. Correspondingly, the damping coefficients are assumed as $c_1 = c_2 = c_3 = c_4 = 0.5N/(m \cdot s^{-1})$.

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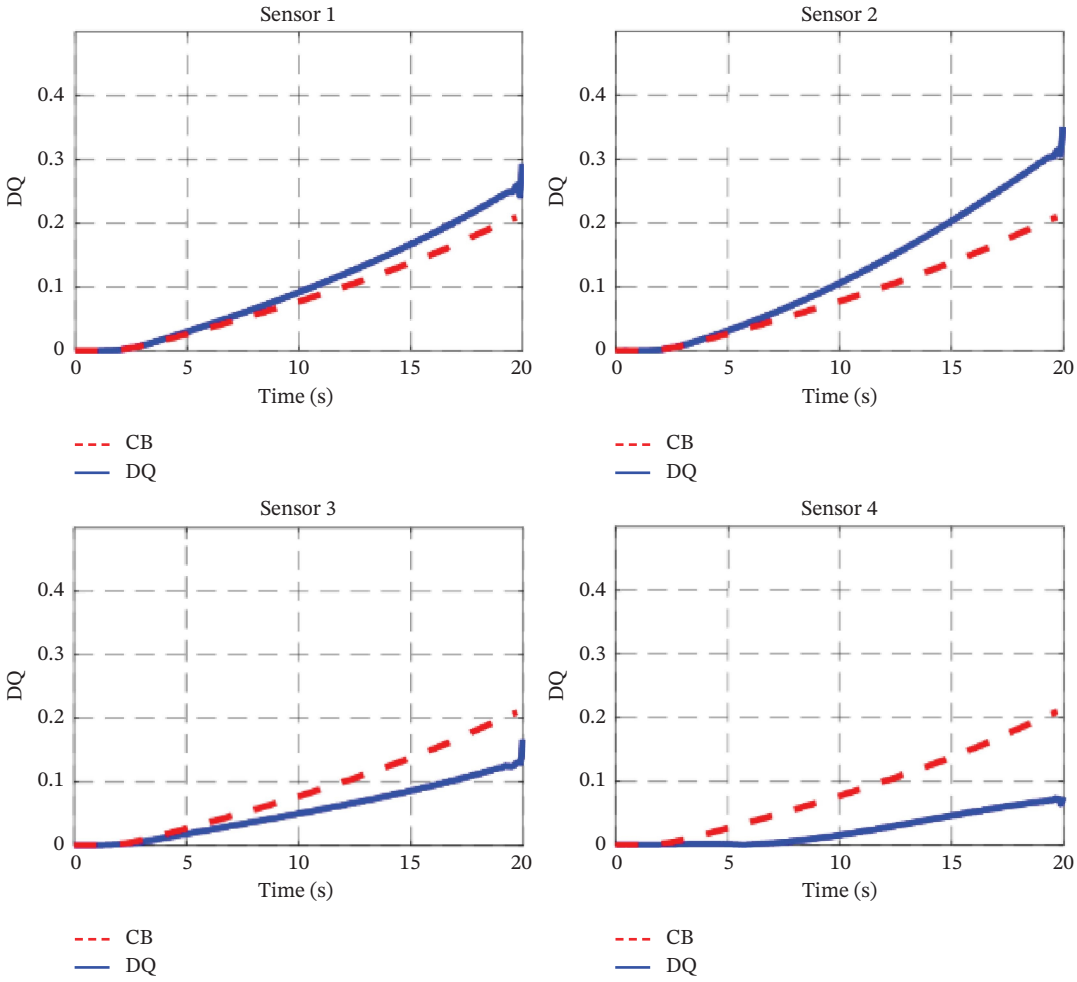


FIGURE 9 | Damage localization results based on the damage index DQ.

The structure is excited by an impulse input case, i.e., the input excitations $F_1(t) = F_2(t) = F_3(t) = 0$; the input excitation $F_4(0) = 300$ and $F_4(t) = 0$ for $t \neq 0$. The resulting acceleration responses are simulated by the Newmark-β method. The simulated accelerations are analyzed to verify the proposed methodology. The sampling frequency and duration of accelerations are 1000 Hz and 20 s. The 1-kHz sampling rate is far higher than the highest natural frequency of the 4-DOF frame, which fully satisfies the Nyquist–Shannon sampling criterion and avoids frequency aliasing while maintaining a manageable data volume. A 20-s signal length delivers a frequency resolution of $\Delta f = 1/T = 0.05$ Hz, which is sufficiently fine to precisely extract modal frequencies and damping ratios, even for closely spaced modes with low damping. No longer time histories are necessary, as 20 s of data supplies complete spectral information for the 4-DOF system. These parameter settings follow standard simulation practices for structural modal identification and are not mandatory fixed values of the algorithm itself. The sliding window length is set to 1024 samples with a 50% overlap ratio; the AR spectrum estimation order is 60. The auto.arima algorithm automatically selects ARIMA orders within $p \in [1, 3]$, $d \in [0, 1]$, $q \in [0, 2]$. All unified ACMD hyperparameters follow the configurations listed in Table 1.

4.1.2 | Results of Time-Varying Modal Parameter Extraction

The acceleration response of the first floor undergoes sliding-window AR power spectrum estimation, producing the time–frequency distribution shown in Figure 3. Evident variations in modal energy levels are observed, where brighter regions indicate higher energy density—specifically, time–frequency ridge lines. By detecting these ridge lines within high-energy regions, instantaneous frequencies are extracted to provide initial center frequencies for ACMD. After isolating the target mode, the corresponding IMFs are subtracted from the original acceleration signals. The residual signals are then reused for subsequent initial center frequency determination and IMF decomposition.

The proposed method successfully identified four modes, including their IMFs and instantaneous frequencies. Figure 4 compares the theoretical frequencies with those identified by the proposed method, demonstrating high consistency except for those at both ends of time. In addition, the time-varying mode shapes identified from IMFs are also presented in Figure 5.

The advantage of the proposed identification method for initial instantaneous frequencies is illustrated by comparison with the results obtained using poorly chosen initial frequencies. Figure 6

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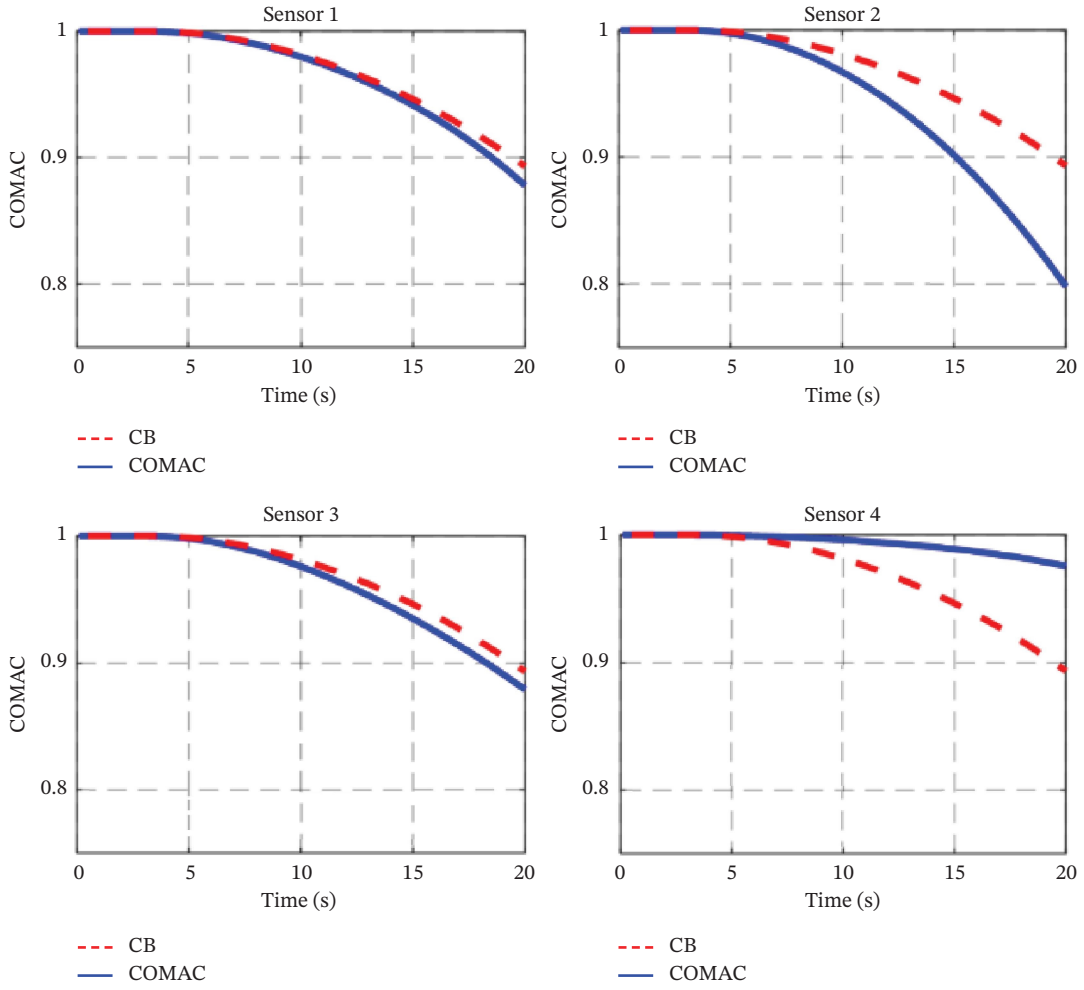


FIGURE 10 | Damage localization results based on the damage index COMAC.

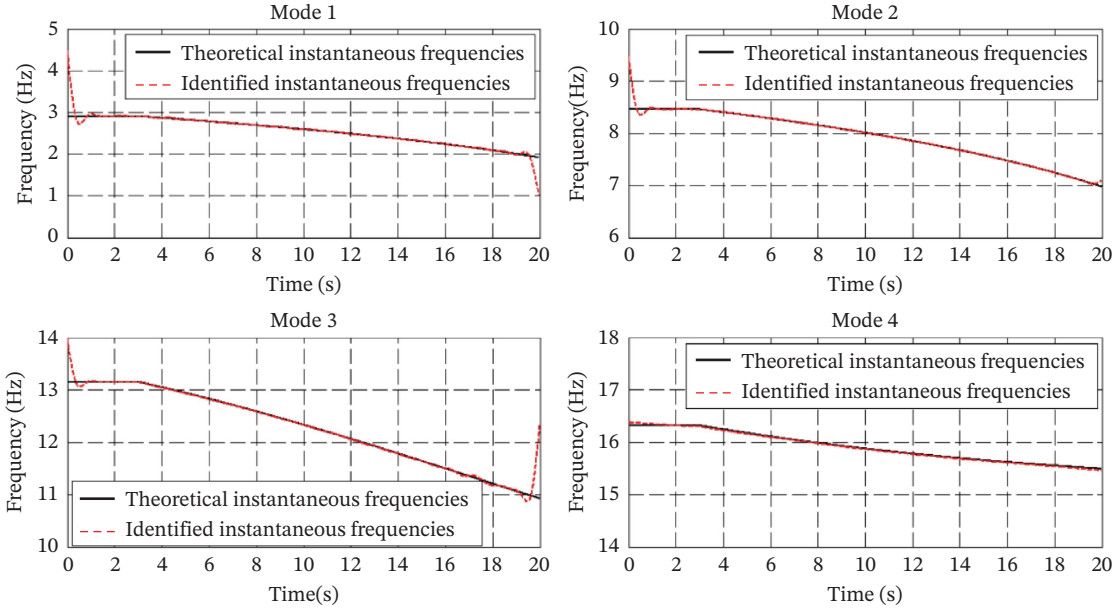


FIGURE 11 | Identified instantaneous frequencies under SNR of 10 dB.

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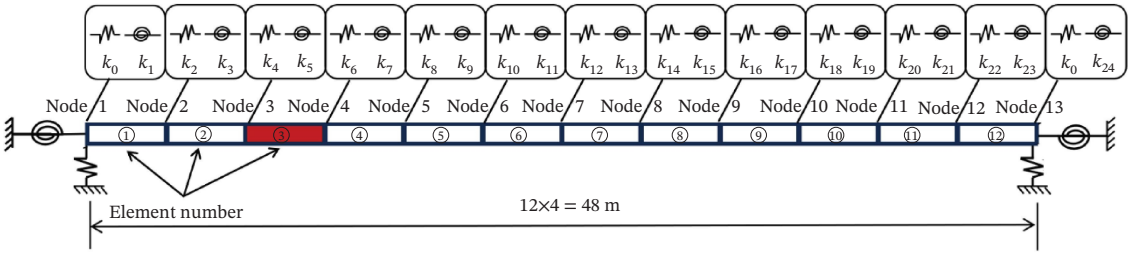


FIGURE 12 | Model of the simply supported beam.

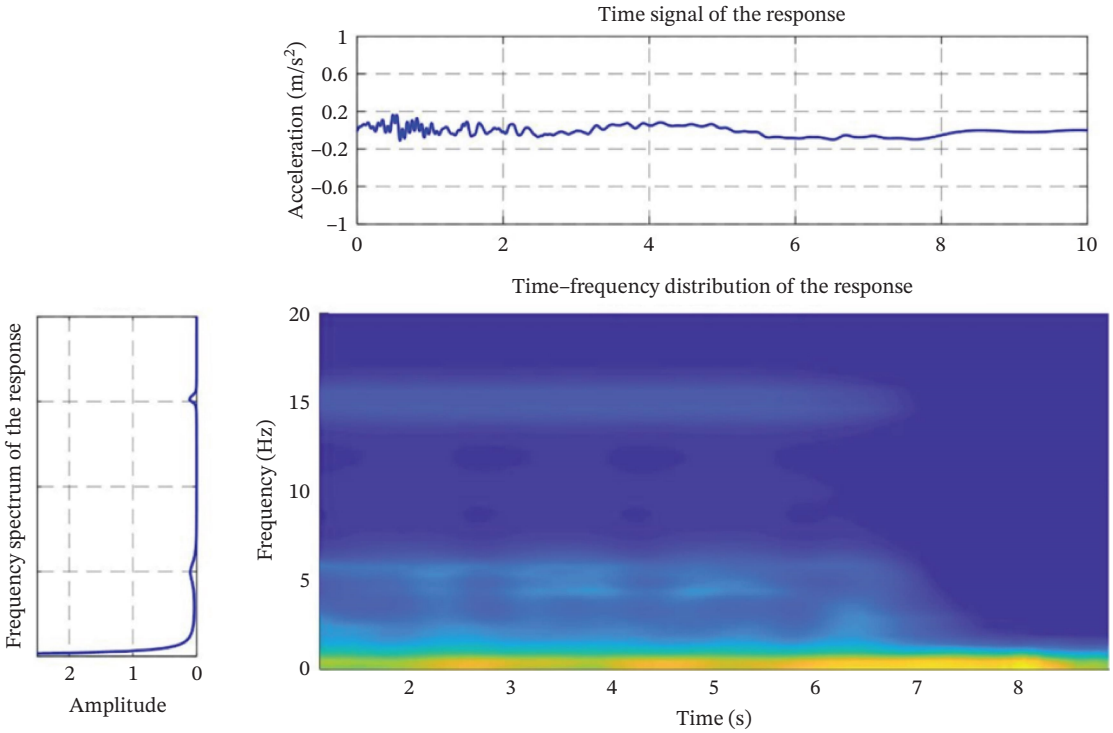


FIGURE 13 | Time–frequency distribution of the acceleration of node four.

shows the instantaneous frequencies identified using poorly chosen initial frequencies. The time–frequency distribution and ridge extraction are obtained by the STFT method. Figure 7 shows the identified instantaneous frequencies by well-chosen initial frequencies. The time–frequency distribution and ridge extraction are realized by the proposed sliding-window AR power spectrum method. Obviously, the poorly chosen initial frequencies will lead to significance errors for the identified instantaneous frequencies by ACMD.

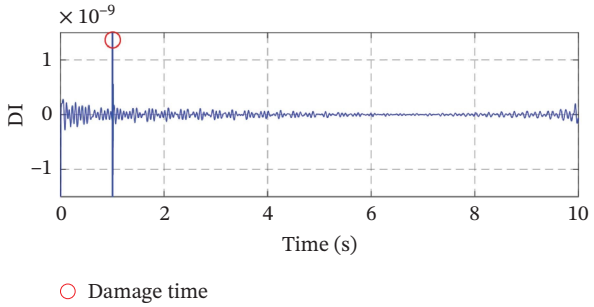


FIGURE 14 | Damage time localization results based on the damage index.

4.1.3 | Analysis of Damage Indices DI and DQ

The extracted IMFs are fitted using the ARIMA model to obtain model residuals, which are then used to construct the DI for identifying the exact time of damage occurrence in the structure. As shown in Figure 8, the DI values exhibit a sharp change at 3 s, indicating the onset of time-varying structural damage.

The DI DQ, based on instantaneous frequencies and mode shapes, is then constructed, and the CB value for each time window is calculated, as shown in Figure 9. This figure clearly shows damage occurring near sensors 1 and 2. For sensors 3 and 4, which are farther from the damage location, the DI DQ does not exceed the CB. The damage location method is compared with the Coordinate Modal Assurance Criterion (COMAC) method. COMAC is defined as

$$\mathrm{COMAC}_s = \frac{\left(\sum_{i=1}^{N}\left|\left(\varphi_s^{(i)}\right)_{\mathrm{ref}}\varphi_s^{(i)}\right|\right)^2}{\sum_{i=1}^{N}\left(\varphi_s^{(i)}\right)_{\mathrm{ref}}^2\sum_{i=1}^{N}\left(\varphi_s^{(i)}\right)^2}, \tag{28}$$

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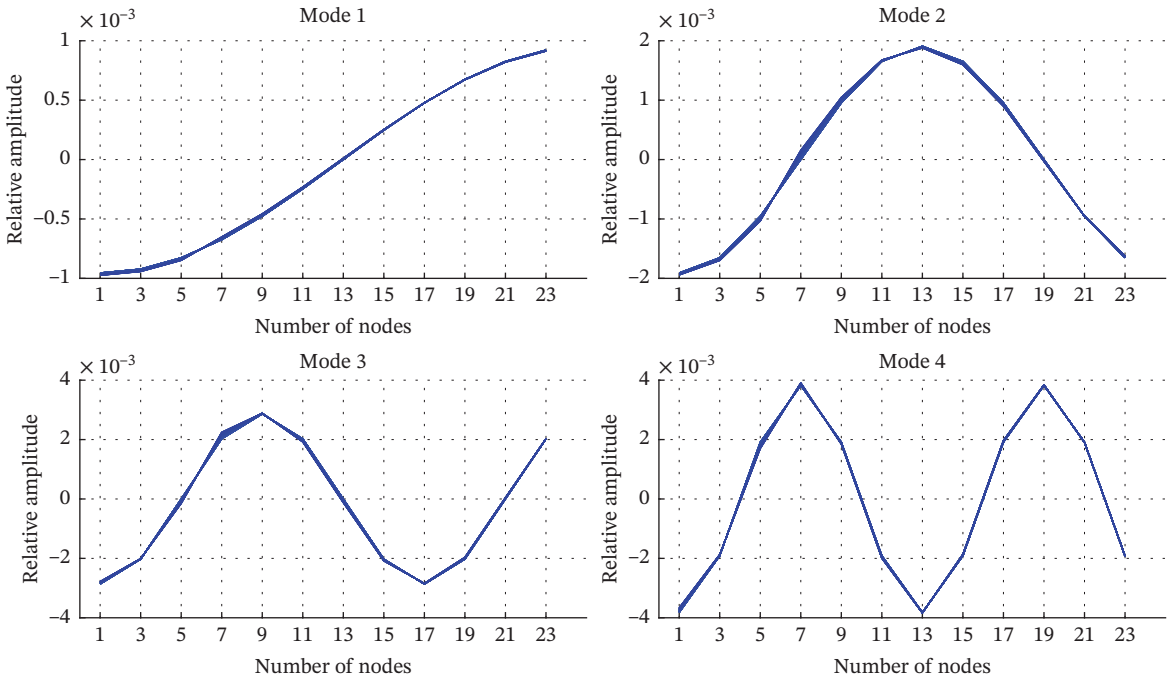


FIGURE 15 | Mode shapes of the simply supported beam.

where $\varphi_s^{(i)}$ is the i-th mode shape; and the subscript "ref" represents the reference mode shape. Figure 10 shows the damage localization results based on the DI COMAC. The proposed DI DQ can accurately localize the damaged sensor 1 and sensor 2. However, the DI COMAC mistakenly identified the damage location as sensor 4. Damage location analysis for the four sensors confirms the effectiveness of the proposed method in identifying damage locations.

4.1.4 | Influence of Measurement Noise

To evaluate the noise robustness of the proposed method, an identification test is conducted under a severe noise condition with SNR = 10 dB. This SNR level represents a challenging field monitoring environment with intensive background disturbances, which serves as a critical limiting case for vibration-based damage detection. As shown in Figure 11, the instantaneous frequencies extracted by ACMD maintain good consistency with theoretical values even under such heavy noise interference. This confirms that ridge detection and subsequent modal decomposition are robust to practical levels of measurement noise. Since accurate time-varying modal parameters serve as the basis for constructing the DI, the temporal detection module can still accurately capture the occurrence time of damage, verifying the favorable noise resistance of the proposed framework.

4.2 | Simulation of a Simply Supported Beam

4.2.1 | Simulation Model and Damage Scenario Setup

To further validate the proposed method's effectiveness, a simply supported beam (Figure 12) is employed, which is modeled using 12 homogeneous Euler–Bernoulli beam elements with the following properties. The cross-section of the beam is 0.2×0.35 m. The Young's modulus is $E = 3.4$ GPa. The density is $\rho = 2800\,\mathrm{kg/m^3}$. The stiffness of the transverse coupling spring K_s is 2.7×10^{10}N/m. The stiffness of the torsion spring K_r is 2.1×10^{9}N/rad. The length of a beam element is 4 m. Two moving forces applied to the beam are assumed to be

$$\begin{cases} P_1(t) = 1600(1 + 0.1\sin \pi t + 0.05\sin 30\pi t)N, \\ P_2(t) = 1600(1 - 0.1\sin \pi t + 0.05\sin 30\pi t)\mathrm{N}. \end{cases} \tag{29}$$

The moving forces traverse the bridge from left to right at a constant speed of 7.5 m/s, passing through all sections. The system employs Rayleigh damping with 0.03 damping ratios for the first two modes. Data acquisition uses a 1000-Hz sampling frequency over a 10-s duration. A window length of 1024 samples with a 75% overlap ratio is adopted to track the fast-varying moving-load responses; the AR spectrum estimation order is set to 60. The ARIMA orders are autoselected within the ranges defined in Table 1, and the standard ACMD parameter settings are utilized.

To simulate the damage conditions at multipositions of the simply supported beam, the stiffness of element 3 is reduced by 25%. In addition, the stiffnesses k_4, k_5, k_6, and k_7 are time-varying. When $t \le 1$s, $k_4 = k_6 = 2.7 \times 10^{10}$N/m, and $k_5 = k_7 = 2.1 \times 10^{9}$N/rad. When $1 < t \le 10$s, $k_4 = k_6 = K_s \times (1 - 2.78(t-1) \times 10^{-5})$N/m and $k_5 = k_7 = K_r \times (1 - 2.78(t-1) \times 10^{-5})$N/rad. The time-varying stiffness is used to simulate the occurrence of progressive damage to the structure at element 3.

4.2.2 | Framework Application and Damage Identification Results

To intuitively display the time–frequency characteristics of vibration signals under moving loads, the time–frequency distribution contour of the acceleration response at node 4 is presented, as shown in Figure 13.

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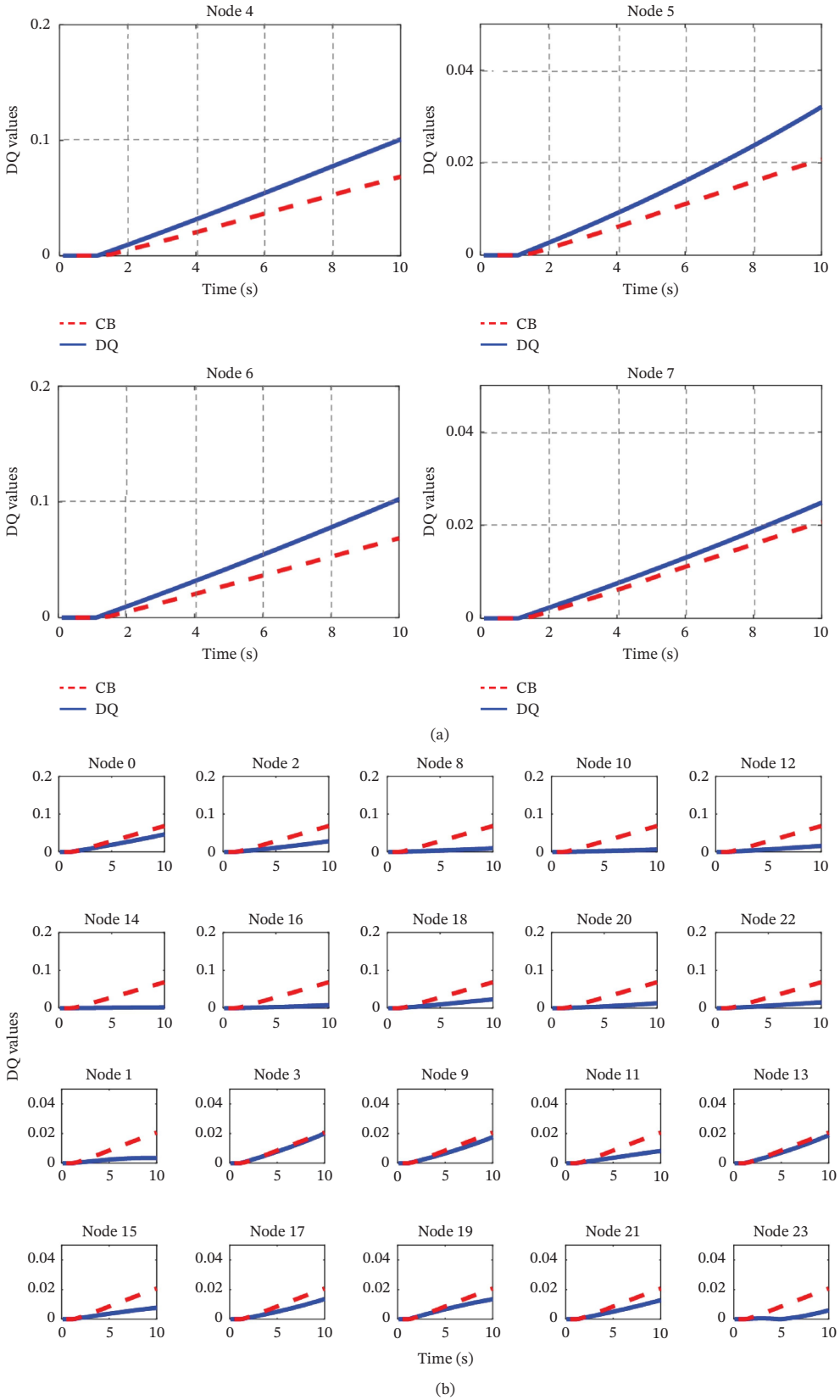


FIGURE 16 | Damage localization results based on the damage index DQ. (a) DQ values of the nodes around the third element. (b) DQ values of the remaining nodes.

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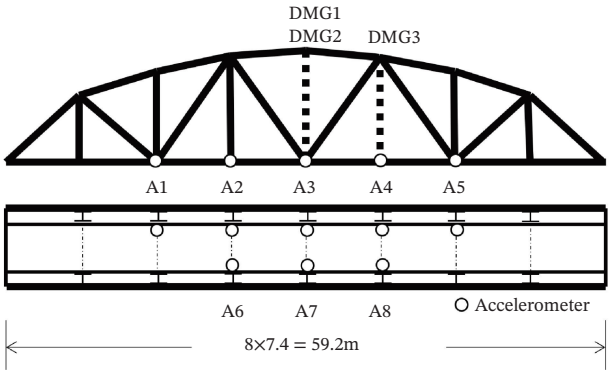


FIGURE 17 | Sensor layout of the actual steel truss bridge.

TABLE 2 | Damage scenarios.

Scenario	Description
INT	Intact bridge (undamaged state)
DMG1	Half cut in vertical member at midspan
DMG2	Full cut in vertical member at midspan
RCV	Recovery of the cut member
DMG3	Full cut in vertical member at 5/8th-span

ARIMA models are fitted to each IMF obtained using the ACMD method to capture their transient characteristics. The damage indicator DI is subsequently constructed based on these model residuals. As shown in Figure 14, the DI values exhibit a sharp change at $t = 1$ s, consistent with the actual damage event. This result validates the effectiveness of the DI indicator for damage detection.

Next, sliding-window PCA processing is applied to the IMFs to obtain the corresponding time-varying mode shapes, as illustrated in Figure 15. After obtaining all instantaneous modal parameters, the damage indicator DQ along with the CB are constructed to identify the specific location of progressive damage in the simply supported beam.

As depicted in Figure 16, the DQ values for nodes 4, 5, 6, and 7 near the third damage element are significantly higher than the CB. This suggests that the vicinity of the mentioned nodes is likely the progressive damage area in the simply supported beam. The continuous increase in DQ values reflects a trend of gradual expansion of damage over time, which is consistent with the manifestation of progressive damage in the simply supported beam.

4.2.3 | Influence of Moving Load Speed and Excitation Variability

The proposed method does not require prior knowledge of excitation characteristics, as it operates in an output-only framework. For moving-load speed, the ridge-guided ACMD extracts instantaneous frequencies that are inherent structural properties, largely independent of the input amplitude or speed, as long as the response is adequately sampled (satisfying the Nyquist criterion). This is confirmed by the simply supported beam results presented in Section 4.2.2, where the DI consistently captures the damage onset at $t = 1$ s under the prescribed moving-load condition. For excitation variability, since the method relies on time–frequency ridges of the output response rather than input–output transfer functions, variations in excitation form and amplitude within typical ambient vibration ranges (e.g., wind, traffic, and microtremors) do not compromise the damage detection results. However, we acknowledge that extreme nonstationary events such as strong earthquakes lie beyond the current scope.

5 | Experimental Validation

This section examines the capability of the proposed diagnostic framework to achieve accurate spatiotemporal damage identification under complex conditions through a case study of ambient vibration monitoring on an actual steel truss bridge.

5.1 | Bridge Description and Experimental Data

This study employs experimental data from an actual bridge, as reported by Chang and Kim [35]. The bridge has a span of 59.2 m and a width of 3.6 m. Figure 17 illustrates the sensor arrangement, including eight accelerometers A1–A8 installed at specific damage locations on both sides of the bridge. Accelerations under ambient excitations are recorded. The sampling frequency is 200 Hz, with a duration of 50 s per test condition. The 50-s recording duration is selected to ensure adequate frequency resolution while maintaining computational efficiency. At a sampling rate of 200 Hz, a 50-s dataset delivers a frequency resolution of $\Delta f = 1/T = 0.02$ Hz, which is fine enough to distinguish the bridge's closely spaced vibration modes. Shorter time histories would blur adjacent modal peaks, while longer records bring limited additional improvement in decomposition accuracy and merely raise computational cost. Thus, this duration is adopted as a practical, efficient parameter setting. A window length of 1024 samples with a 50% overlap ratio is adopted, and the AR spectrum estimation order is set to 60. The ARIMA and ACMD hyperparameters comply with the unified configuration rules summarized in Table 1.

All vibration signals of the steel truss bridge are collected under ambient environmental excitation, consisting of stochastic broadband excitations induced by wind loads, passing vehicle traffic, and ground microtremors. Such inputs are nonstationary with continuously time-varying amplitudes. Although fluctuations in excitation strength will change the energy magnitude of the signal spectrogram and slightly alter the peak amplitude of

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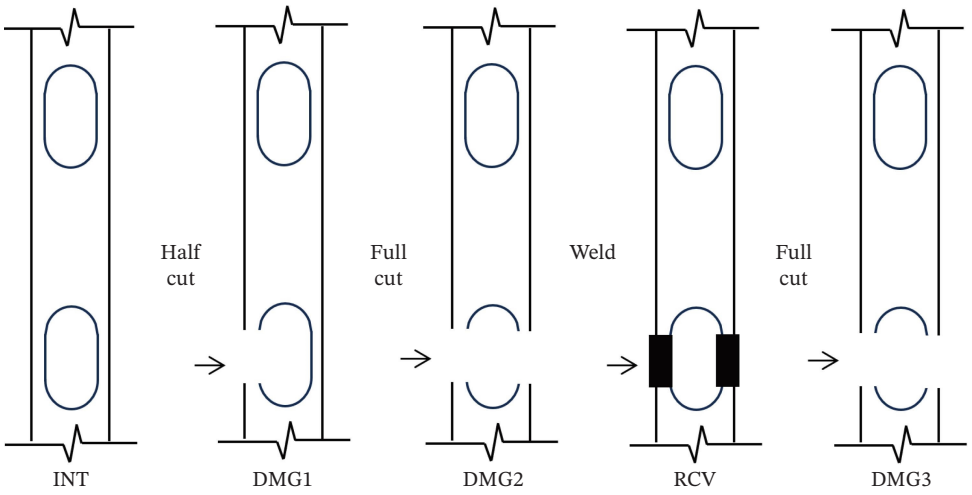


FIGURE 18 | Damage conditions of the structural components.

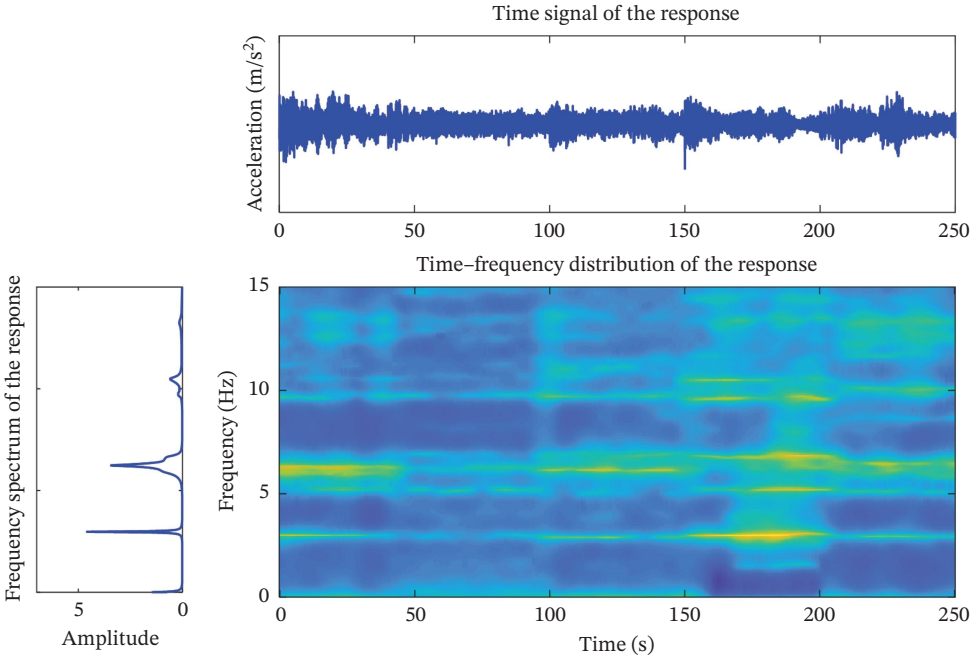


FIGURE 19 | Time-frequency distribution of the acceleration of node NO. A1.

extracted time–frequency ridges, the natural frequencies and mode shapes identified by the ridge-guided ACMD are intrinsic linear structural properties, which remain stable and insensitive to excitation amplitude variations. The iterative refinement mechanism of ACMD can eliminate minor disturbances brought by variable excitation, so the modal decomposition results will not be distorted by normal ambient fluctuations. Consequently, the variations of DI and DQ indices observed in subsequent analysis mainly originate from local stiffness degradation (structural damage), rather than changes in environmental excitation.

Five scenarios of the bridge are introduced in Table 2 and Figure 18. The acceleration responses from all five scenarios were concatenated, resulting in a 250-s dataset. All five scenarios were tested under ambient excitations. Time–frequency analysis was then applied to extract time-varying modal parameters, enabling the identification of damage locations and their respective occurrence times.

5.2 | Results and Validation of Spatiotemporal Damage Localization

Figure 19 illustrates the time–frequency distribution obtained through sliding-window AR power spectrum analysis for the acceleration of node NO. A1. By applying time–frequency ridge extraction, high-energy instantaneous frequencies are identified and subsequently used as initial inputs for the ACMD algorithm. This ensures that targeted modes can be extracted by the ACMD algorithm, thereby enhancing its accuracy and efficiency in practical applications. The proposed method successfully extracts four IMFs and their corresponding instantaneous frequencies, and the instantaneous frequencies are shown in Figure 20. The identified instantaneous frequencies in Figure 20

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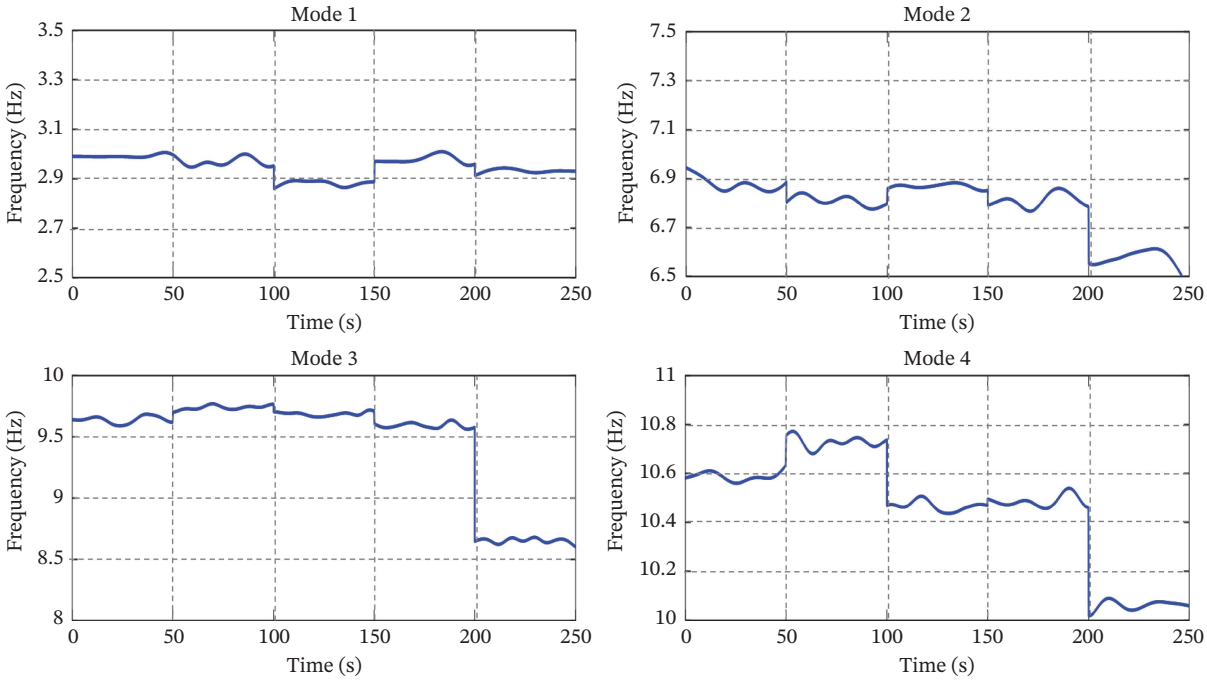


FIGURE 20 | Identified instantaneous frequencies based on ACMD.

exhibit variations across different damage scenarios that may appear non-intuitive at first glance. This can be explained by the mode-dependent sensitivity of frequency shifts to local stiffness reductions. It should be noted that pure structural stiffness loss cannot physically raise natural frequencies. Specifically, the frequency change of a given mode due to localized damage is proportional to the modal strain energy density at the damage location. For DMG2, the introduced damage is confined to regions where certain modes have low strain energy (near mode shape nodes of modes 3 and 4), resulting in negligible frequency variation; the minor upward deviation of frequency estimates is merely caused by environmental variability and identification uncertainty. For DMG3, as damage extends into areas that are more sensitive to these modes, more pronounced frequency drops are observed. This interpretation is consistent with the spatial distribution of the introduced damage and the strain energy maps of the corresponding modes.

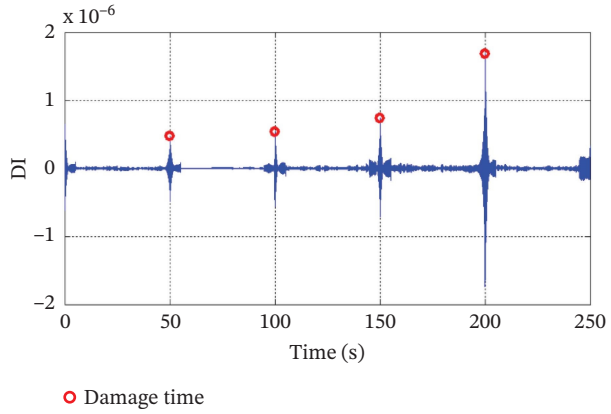


FIGURE 21 | Damage time localization results based on the damage index.

The extracted IMFs are fitted with ARIMA models to obtain residuals, from which the DI is constructed to pinpoint the exact damage times. As shown in Figure 21, abrupt changes in DI values at 50, 100, 150, and 200 s clearly mark the identified damage instances.

Subsequently, instantaneous mode shapes are derived, as shown in Figure 22. The damage location index DQ, constructed from time-varying modal parameters, and the CB are presented in Figure 23. It can be observed from Figure 23 that the DQ values at the third sensor location exceed the CB from 50 to 150 s, indicating that the damage occurred near the third sensor during this period. The DQ values remain above the damage threshold CB from 150 to 200 s. Therefore, the third sensor location is identified as abnormal. The DQ values at the fourth sensor location exceed CB from 200 to 250 s, indicating that the damage occurred near the fourth sensor location during this period. The high consistency between the damage development revealed by this method and the actual scenario validates its accuracy and reliability. The results show that the identified damage time and locations are consistent with the actual scenario. Therefore, the method's effectiveness for practical application is verified.

5.3 | Discussion on Sparse Sensing Conditions

The three validation examples presented above adopt relatively dense sensor layouts, with measuring points arranged close to preset damage zones. Nevertheless, practical on-site structural health monitoring systems frequently adopt sparse sensor networks due to budget limits and installation constraints.

The spatial damage localization module relies on the spatial variation of modal characteristics (frequency shifts and mode shape distortions) to quantify damage positions. Theoretically, the method's sensitivity to minor local stiffness degradation declines once the sensor spacing exceeds half the characteristic wavelength of target modes. In this case, subtle spatial

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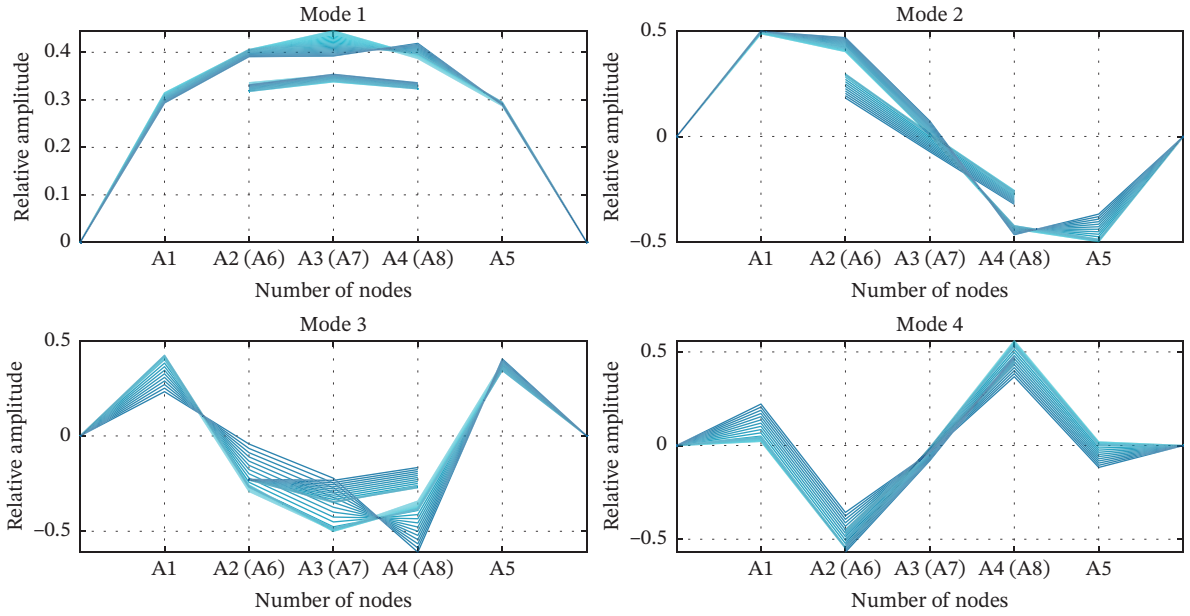


FIGURE 22 | Identified time-varying mode shapes.

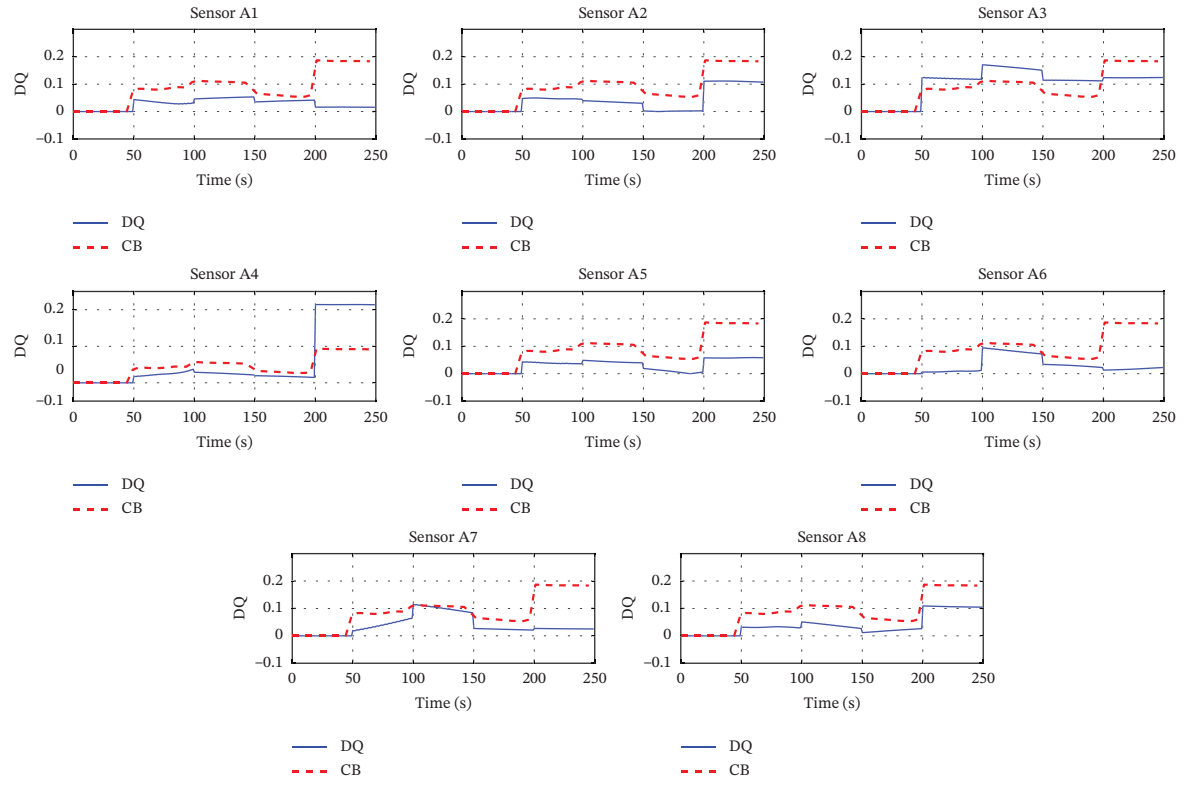


FIGURE 23 | Damage localization results based on the damage index DQ.

fluctuations of structural vibration responses cannot be fully sampled, leading to spatial smoothing of the DQ index and reduces localization precision. In contrast, the global damage detection module based on the ARIMA-residual-based DI only requires single-point time-series data, thus exhibiting strong robustness regardless of sensor sparsity.

Two feasible auxiliary strategies can be combined with the proposed framework to mitigate resolution loss under sparse sensing. On the one hand, optimal sensor placement algorithms can allocate limited sensors to regions with high modal strain energy to maximize damage sensitivity. On the other hand, finite-element-model-driven mode shape expansion reconstructs dense full-field deflection data from sparse measurements, recovering sufficient spatial resolution for precise damage localization. Quantitative parametric studies and sparse-sensing experimental verification to quantify the method's performance threshold are left as future research. Furthermore, instantaneous mode shapes

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in the present study are obtained only at instrumented sensor locations; no interpolation was applied to recover unmeasured degrees of freedom. Incorporating finite element–based mode shape expansion or virtual sensing techniques to reconstruct denser deflection fields from sparse measurements is a promising direction for enhancing spatial localization accuracy and is left for future work.

5.4 | Computational Efficiency Analysis

The proposed two-stage diagnostic pipeline integrates sliding-window segmentation, AR spectrum estimation, ridge-guided ACMD modal decomposition, PCA feature reduction, and ARIMA time-series modeling.

The overall runtime is primarily influenced by (i) the number of sensors, (ii) the total data length, (iii) the number of windows (which determines how many times the modal decomposition is repeated), and (iv) algorithm parameters such as window length, overlap, model order, and number of retained modes.

All computations for the validation cases were implemented on a conventional desktop PC (Intel Core i7, 3.2 GHz, 16 GB RAM) without GPU or parallel computing acceleration. For every case studied in this paper, the complete end-to-end processing procedure, from raw signal input to final DI output, was completed within several seconds (typically 5–15 s, depending on the dataset size). Such low latency demonstrates that the presented framework can support near-real-time damage detection for small- and medium-scale structures. For large-scale engineering structures equipped with dense sensor networks and long-duration continuous monitoring data, further algorithmic optimization and parallel window processing will be investigated in follow-up research to further reduce computational latency.

6 | Conclusions

In this study, a novel two-stage diagnostic framework for progressive damage identification in time-varying structures has been proposed. The framework is characterized not merely by the application of the ACMD but by the targeted resolution of its practical limitations in the context of structural health monitoring, coupled with the development of its outputs for comprehensive damage diagnosis. It provides a practical solution for damage monitoring and preventive maintenance of time-varying structures in practical engineering. The conclusions are summarized as follows:

1. The introduction of a sliding-window AR power spectrum-based time–frequency ridge extraction mechanism effectively overcomes the sensitivity of ACMD to initial instantaneous frequency estimation. This method adaptively captures high-energy modal frequency ridges, providing reliable initial values without requiring prior knowledge or extensive tuning. The improvement significantly enhances the robustness and accuracy of modal decomposition for nonstationary structural responses.
2. Beyond modal decomposition, the extracted IMFs are further utilized to construct two complementary damage indices. The DI, derived from the residual sequences of ARIMA models fitted to the IMFs, enables sensitive and accurate identification of damage occurrence time. The DQ index, which integrates time-varying frequency shifts and local mode shape distortions, allows for precise spatial localization of progressive damage. Together with a time-varying CB, the framework offers a statistically grounded approach to damage tracking and localization.
3. The validity and reliability of the proposed framework have been rigorously verified through numerical simulations and experimental studies. The simulation cases include a time-varying 4-DOF frame structure and a simply supported beam under moving loads. These cases demonstrate that this method can accurately detect and locate damage caused by progressive stiffness changes. Moreover, when applied to real environmental vibration data from a steel truss bridge, it has also been confirmed to have practical feasibility and diagnostic accuracy in complex operating conditions.

Despite the promising diagnostic results achieved in this study, the proposed framework still carries several inherent limitations. First, the method was developed for linear structural systems and weakly stationary excitations, and its reliability under strong structural nonlinearities or highly nonstationary load events remains unvalidated. Second, the spatial resolution required by the DQ damage localization index restricts performance; localization precision will deteriorate when the sensor network is sparsely arranged. Third, all validation tests consider only a single progressive damage location, while scenarios with multiple concurrent damages distributed at different locations are not investigated in the current work.

To overcome the above limitations, future research will extend the pipeline to handle nonlinear structures and strongly nonstationary excitations. Additional efforts will be devoted to improving the method's robustness under sparse sensing via optimal sensor layout design or FE-based mode shape expansion. We will also expand the algorithm to identify multiple coexisting damage regions and embed machine learning modules to realize fully automated, real-time online structural health monitoring.

Acknowledgments

This research work was supported by the National Natural Science Foundation of China (Grant No. 52308326), the Natural Science Foundation of Hebei Province (Grant No. E2023202101, E2026202176), the Natural Science Foundation of Fujian Province (Grant No. 2025J08030), the Research Starting Funding of Fuzhou University (XRC-24053), and the Transportation Science and Technology Development Plan Project of Tianjin (Grant No. 2024-G13).

Funding

This study was funded by the National Natural Science Foundation of China, 52308326; Natural Science Foundation of Hebei Province, E2023202101 E2026202176; Natural Science Foundation of Fujian Province, 2025J08030; Research Starting Funding of Fuzhou University, XRC-24053; and Transportation Science and Technology Development Plan Project of Tianjin, 2024-G13. Open access publishing facilitated by Politecnico di Milano, as part of the Wiley - CRUI-CARE agreement.

Conflicts of Interest

The authors declare no conflicts of interest.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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